Nonlinear Diffusion-Enhanced Feature Representation and Matching for Block Adjustment of Multiscale Optical Satellite Imagery

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***Abstract*—Deep learning-based feature matching methods have been actively explored for satellite image bundle block adjustment, leveraging their inherent robustness to large geometric distortions and radiometric differences—key challenges in multisource optical image processing. However, their practical utility remains limited by two critical bottlenecks: poor adaptability to extreme scale variations and prohibitive computational costs for large-format satellite data. To address these limitations, we propose a learning-based feature representation method enhanced by nonlinear diffusion filtering, with two targeted innovations: 1) nonlinear diffusion filtering with terrain-adaptive parameters is integrated into a novel image tiling strategy, which preserves local feature integrity while enabling consistent correspondence across heterogeneous satellite data sources and 2) a top–down pyramid construction mechanism that incorporates local continuity constraints and an adaptive matching strategy selection protocol optimizes scale space exploration efficiency while safeguarding matching quality. Experiments on Earth observation and Martian image datasets validate the method's superiority: it achieves the highest matching success rate (97.25%) and the highest BBA accuracy (1.65 pixel) among competing approaches, alongside competitive efficiency. This performance advantage is particularly pronounced under extreme scale differences and challenging imaging conditions, confirming its suitability for high-precision remote sensing applications.**

***Index Terms*—Bundle block adjustment (BBA), image matching, nonlinear diffusion filtering, optical satellite images.**

I. Introduction

BUNDLE block adjustment (BBA) is a cornerstone process in satellite remote sensing, enabling high-precision geolocation and mapping [1]. A widely adopted approach to BBA involves image matching, which establishes spatial geometric constraints to enhance both geometric consistency and positioning accuracy [2]. With the rapid growth in the volume and diversity of optical satellite imagery from Earth observation and deep-space exploration missions, the demand for BBA based on multisource optical imagery has increased significantly [3]. However, variations in acquisition time, viewing angles, and sensor configurations, combined with dynamic surface changes, pose significant challenges for multisource image matching [4]. These challenges include severe geometric distortions, nonlinear radiometric discrepancies, and large-scale variations [5].

Satellite image matching methods can be broadly classified into three categories. Region-based methods, such as normalized cross correlation (NCC), utilize local gray-scale and texture similarities for matching [6]. These methods are effective in scenarios with stable features and minimal geometric distortions but are highly sensitive to radiometric differences and texture deficiencies. Feature-based methods, on the other hand, extract distinctive features such as corners and edges. The scale-invariant feature transform (SIFT) algorithm is a classic example, offering robustness through rotation and scale invariance [7]. However, its high computational complexity and reliance on manually designed features limit its applicability in scenarios involving nonlinear radiometric discrepancies. The speeded-up robust features (SURFs) algorithm improves computational efficiency using integral images but sacrifices some feature stability [8].

In recent years, learning-based methods have emerged as transformative tools in remote sensing image processing, showcasing significant potential in addressing the challenges of multisource image matching [9]. These methods leverage data-driven feature extraction capabilities to handle geometric distortions and radiometric discrepancies more effectively than traditional approaches [10]. Algorithms such as the SuperPoint–Superglue framework, the lightweight LightGlue, and end-to-end networks like Local Feature TRansformer (LoFTR) have demonstrated remarkable adaptability to the inherent complexities of multisource imagery [11], [12], [13], [14]. Their ability to learn complex feature representations automatically positions them as promising solutions for the increasingly demanding applications in modern remote sensing.

Several studies have explored the application of learning-based methods to remote sensing imagery. For instance, Morelli et al. [15] implemented these methods on large-frame remote sensing images by processing satellite imagery in blocks and in parallel, though geographic positioning infor-

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mation was not fully considered. Similarly, Cheng et al. [16] proposed the MSRSI-TPMF framework, which unified the matching of multisource satellite images and integrated heterogeneous deep learning-based matching methods. However, this framework did not address the subsequent application of multisource optical satellite image matching in regional network adjustment. Furthermore, Song et al. [17] and Huang et al. [18] investigated the performance of deep learning methods in matching de-orbited high-resolution stereo satellite images, with results demonstrating superior adjustment accuracy and terrain reconstruction quality compared with traditional methods [19].

Despite these advancements, challenges remain. One of the most critical issues is the robustness of deep learning-based methods to scale variations, which significantly impact feature representation. Ground objects often exhibit substantial differences in spatial structure, texture details, and contextual information across images captured at varying resolutions. Consequently, deep neural networks may generate inconsistent feature responses for the same object observed at different scales, compromising the consistency of feature representations for corresponding points [20]. This limitation is particularly pressing in photogrammetric bundle adjustment with multisource remote sensing imagery, where scale variation remains a fundamental challenge.

To address scale differences, existing methods can be divided into two categories. The first category focuses on reducing scale differences through downsampling and image alignment techniques, such as image segmentation, resolution unification, or orthorectification. For example, Morelli et al. [15] segmented and downsampled images to maintain input consistency for neural networks, while Cheng et al. [16] unified image resolution through orthorectification to reduce scale and rotation differences. However, these methods often lose contour and texture details in high-resolution images, reducing the number of features. In addition, scale differences may persist due to rational polynomial coefficient (RPC) errors and rough digital elevation model (DEM). The second category achieves scale invariance by constructing multiscale feature spaces. For instance, the SIFT algorithm uses a multiscale Gaussian pyramid combined with the difference of Gaussian to detect key points and achieve scale normalization. Similarly, the KAZE algorithm introduces nonlinear diffusion filtering to construct a nonlinear scale space, preserving edge and detail features more effectively [21]. AKAZE further accelerates the solution of nonlinear diffusion equations using the fast explicit diffusion (FED) method, significantly improving computational efficiency [22].

Both approaches aim to eliminate scale differences and achieve optimal matching when the matching scale is unknown. However, in multisource remote sensing satellite image matching, the positioning information of the images provides a scale prior, allowing the sampling direction of image pairs to be directly determined. This eliminates the need to construct consistent pyramid scale spaces for image pairs. Instead, a continuous scale space can be constructed for high-resolution images alone, reducing computational complexity and improving matching efficiency. In addition, for challenging

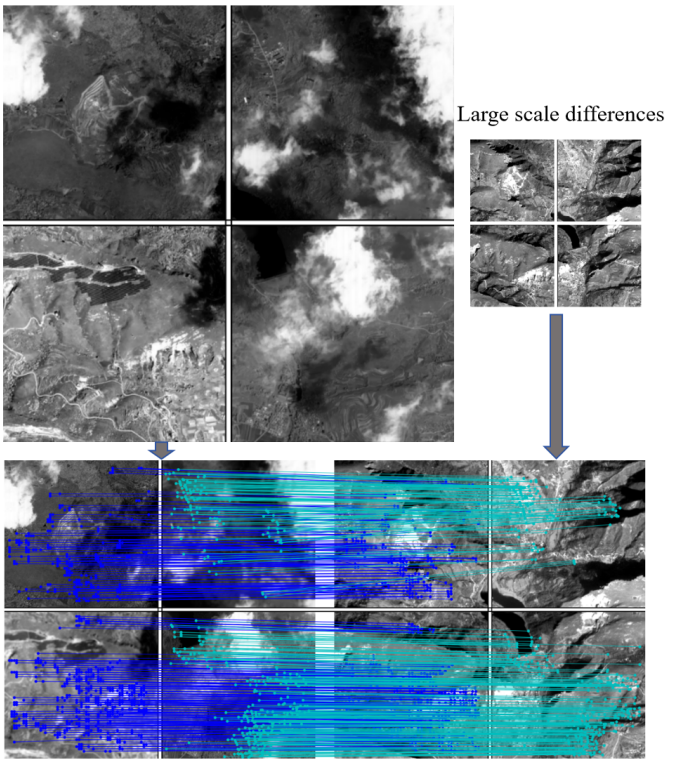

Fig. 1. Corresponding point matching between images with significant scale differences using our proposed methods.

scenarios such as geometric distortion, weak texture, and poor lighting conditions—especially in deep-space remote sensing—combining deep learning methods with nonlinear filtering algorithms can further enhance feature extraction robustness.

To this end, this article proposes a learning-based multiscale satellite image feature representation and matching method enhanced by unilateral nonlinear diffusion filtering, specifically designed for block adjustment of multiscale optical satellite images. This method constructs a scale pyramid for high-resolution images, uses positioning prior information to determine the initial sampling scale, and builds a continuous sampling space from top to bottom to optimize the matching scale. Nonlinear diffusion filtering with adaptive κ values effectively preserves texture and edge features, improves feature extraction accuracy and stability, and accurately regresses feature points in the multiscale space to the original resolution, significantly enhancing matching accuracy and robustness. In Fig. 1 illustrates the corresponding point matching between images with significant scale differences using the proposed method.

Specifically, the contributions of this article are listed as follows.

1) We develop a learning-based feature representation framework integrating conventional feature engineering with deep learning techniques, which is specified for matching image correspondences in the BBA.
2) By introducing nonlinear diffusion filtering with terrain-adaptive parameters, we propose a novel image tiling strategy that enables consistent feature correspondence across heterogeneous satellite data sources.

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3) We establish a top–down pyramid construction mechanism with local continuity constraints coupled with an adaptive matching strategy selection protocol. This dual approach optimizes scale space exploration efficiency while ensuring matching quality preservation.

The remainder of the article is structured as follows. First, a thorough review of methods concerning satellite image matching methods is given in Section II. Afterward, in Section III, the proposed framework for learning-based feature matching is explained in detail. Subsequently, experimental results are given and discussed in Section IV. Finally, conclusions are presented in Section V.

II. Related Work

Existing satellite remote sensing image matching mainly adopts two types of strategies: conventional matching methods based on artificially designed features and learning-based matching methods.

A. Conventional Satellite Image Matching Methods

Conventional satellite image matching methods are mainly based on manually designed feature extraction and descriptors, which have been widely used in computer vision and remote sensing image processing. For example, a series of algorithms represented by SIFT [7] use a Gaussian scale space to perform multiscale analysis on images and extract directional histogram features of keypoints and their neighborhoods, thus achieving feature matching that is invariant to scale changes and rotations.

In [23], a preorientation step and geometric constraints were added to the standard SIFT to address the rotation inconsistencies of high-resolution images with strong geometric distortions. In [24], the standard SIFT was modified with an adaptive binning step (AB-SIFT), which dynamically adjusts the spatial and orientation bins based on local contexts of images. Experiments showed that AB-SIFT is more reliable when matching remote sensing images with significant radiometric and geometric distortions. In [25], a hierarchical structure was used to optimize the standard SIFT for global optimization in large-scale aerial photogrammetric matching. In [26], a method that integrates feature-based strategies for keypoint detection and area-based optimization to align image regions was presented to enhance the speed and robustness of multimodal remote sensing image registration. This approach extracts distinctive features for coarse matching using a tailored SIFT algorithm and then employs area-based optimization that minimizes intensity differences between images for fine matching. In [27], the radiation-variation insensitive feature transform (RIFT) was proposed, which is designed to be insensitive to radiation variations by focusing on the relative gradient orientations of image patches. This approach is particularly effective when aligning images from different sensors or taken under varying illumination conditions.

However, the idea based on traditional scale space construction has certain limitations in multisource satellite image matching. First, since multisource satellite images usually have large resolution differences (i.e., scale differences), the performance of traditional methods is significantly reduced when matching across scales. Second, multisource images have differences in acquisition time, sensor characteristics, and imaging angles, resulting in differences in the radiation characteristics, geometric distortion, and texture details of the images, which increases the difficulty of matching. In addition, there are often large flat areas and repeated textures in satellite images, which make the distribution of feature points uneven, affecting the robustness of matching [28]. To address these issues, methods that incorporate nonlinear scale space construction and adaptive feature extraction techniques have been proposed [27], [29], [30]. For example, KAZE [21] employs nonlinear diffusion filtering to construct a scale space, thereby preserving edge and detail information effectively while enhancing performance in natural images. Subsequent improvements, such as Ordóñezet al.'s [31] application of KAZE features to hyperspectral image alignment, and Liu et al.'s [32] scale-adaptive AKAZE extension, enhanced the robustness of KAZE-based features across heterogeneous imaging conditions. Building further on nonlinear scale space principles, Yao et al. [29] introduced CoFSM to address sensor differences in multimodal imaging through adaptive radiometric normalization.

In addition, phase-based matching methods have demonstrated strong adaptability to scale and rotational variations. Histogram of oriented phase congruency (HOPC) [33] represents a significant advancement by leveraging geometric structural properties through phase congruency computation, achieving invariant performance under illumination and contrast changes. Extending this concept, local phase information was systematically utilized for feature extraction in [34], ensuring robustness against intensity variations and noise. Fourier-based correlation combined with phase congruency [35] further enhanced subpixel alignment accuracy under varying illumination conditions. The latest development in this domain involves integrating steerable filters and gradient information with optimized NCC and multiscale analysis [36], enabling robust multimodal registration even in the presence of complex geometric and radiometric distortions.

B. Learning-Based Satellite Image Matching Methods

In the past decade, deep learning technology has made significant progress in the field of computer vision and has also been widely used in feature matching tasks. Matching methods based on deep learning automatically learn discriminant feature representations from data through neural networks, offering enhanced adaptability to complex variations compared with traditional, manually designed descriptors. Early studies mainly use patch-based strategies, focusing primarily on estimating displacement fields or pixelwise shifts between images. For example, in [37], a deep learning-based framework employing a fully convolutional network (FCN) was developed to estimate displacement fields between input image pairs, capturing both global and local spatial transformations. Similarly, in [38], an FCN-based method was presented to extract dense feature representations from input image pairs and estimate the transformation parameters required for alignment. In [39],

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the proposed method extracts spatial and spectral information through two symmetrical weight-sharing convolutional neural network (CNN) channels and uses quadratic polynomial fitting to achieve subpixel alignment accuracy.

However, patch-based methods perform poorly with rotation and scale differences as they rely directly on pixel intensity distributions and lack intrinsic invariance to geometric transformations. In contrast, feature point-based matching methods can better cope with geometric changes under different perspectives. For instance, in [40], a CNN-based approach involves using a pretrained CNN to extract high-level semantic features from input images, which are then matched to determine corresponding points. By extracting corresponding points for transformation, this method is deemed more robust to variations in illumination, noise, and geometric distortions. To remove wrongly matched keypoints, Kim et al. [41] proposed the FMPR-Net, which evaluates the matching probability of pairs using two weight-sharing deep feature extractors. The method is designed for high-resolution satellite images.

Among many feature point-based matching methods, the combination of SuperPoint and Superglue is widely regarded as a representative and effective approach. SuperPoint [11] is an end-to-end feature point detection and descriptor extraction method. It uses self-supervised learning to train a CNN to simultaneously complete key point detection and descriptor generation. On this basis, Superglue [12] uses a graph neural network (GNN) to match the features extracted by SuperPoint. By modeling the relationship between features and global context information, it significantly improves the accuracy and robustness of matching. Superglue can effectively solve the matching difficulties caused by occlusion, lighting changes, and perspective changes. LightGlue [13], as a lightweight optimization of Superglue, adopts sparse feature matching and adaptive layer selection to reduce computational complexity while maintaining comparable matching accuracy, making it more suitable for resource-constrained scenarios. LoFTR [14] proposed a feature matching framework based on Transformer. Through dense feature representation and a cross-layer self-attention mechanism, it can directly perform high-precision dense matching without relying on key point detection. LoFTR performs well in weakly textured and textureless areas and is suitable for matching tasks like those in planetary scenarios. Efficient-LoFTR [42] optimizes LoFTR's Transformer structure with sparse attention and feature downsampling, reducing runtime (RT) by nearly 30% while retaining most of its robustness to weak textures, addressing LoFTR's high computational cost issue.

These deep learning-based matching methods perform well in natural images and standard computer vision tasks, but they still face challenges in matching multisource satellite images [43], [44]. Most deep learning-based methods do not consider large-scale differences during training. When directly applied to satellite images with large differences in resolution, the matching performance will be reduced. Lack of rotation invariance: Satellite images may have large rotation differences due to different imaging angles and orbits. If the matching method is not robust to rotation, the matching effect will be affected. High computing resource requirements: Deep learning methods usually require a large amount of computing resources, and processing high-resolution satellite images will bring a large computing overhead.

To address these challenges, some studies have tried to apply deep learning matching methods to remote sensing image matching and improve them according to the characteristics of satellite images. For example, Zhang and Zhang [45] proposed a method combining deep features and traditional features for high-resolution remote sensing image matching, which improved the matching accuracy. Wang et al. [46] used multiscale deep feature fusion to enhance the robustness of the model to scale changes. However, these methods still cannot completely solve the matching problem under large-scale differences. Therefore, in view of the special needs of multisource satellite image matching, designing a matching method with scale robustness and efficiency has become an urgent issue that needs to be solved.

III. Methodology

In this study, we propose a learning-based feature representation and matching framework with nonlinear diffusion filtering for block adjustment of multiscale satellite images. First, the framework chooses the high-resolution image as the reference image based on its ground sample distance (GSD). Then, the RPC projection transformation is used to calculate the overlap between the reference image and the target images, and the target image with the largest overlap area is selected for matching. We obtain the initial downsampling scale from the GSD and adaptively calculate the parameters of the local pyramid, such as octaves, number of layers, and scale factors. Based on multiscale representation, we propose an adaptive κ-based nonlinear diffusion filtering, which can better preserve useful structures. By integrating the κ value into a dynamic routing mechanism, the framework adaptively switches between the SuperPoint + Superglue (hereinafter referred to as SP + SG) module for high-texture regions and the LoFTR module for weakly textured regions, effectively combining their complementary advantages to significantly enhance the overall matching accuracy of satellite images. The overall workflow of the proposed method is illustrated in Fig. 2.

A. Cross-Scale Overlapping Area Selection

In the process of satellite image matching, it is very important to determine the reference image and the overlap area between the images to be matched. This section provides a detailed description of how to utilize the GSD to select high-resolution satellite imagery as the reference image and employ the RPC model for projection transformations to identify the corresponding overlapping areas of each target image. The process of cross-scale overlapping area selection based on RPC and GSD is illustrated in Fig. 3.

For satellite images, the GSD can be calculated based on the parameters of the satellite sensor and the imaging geometry. In general, the GSD can be calculated as

$$\mathrm{GSD} = \frac{H}{f} \times p \tag{1}$$

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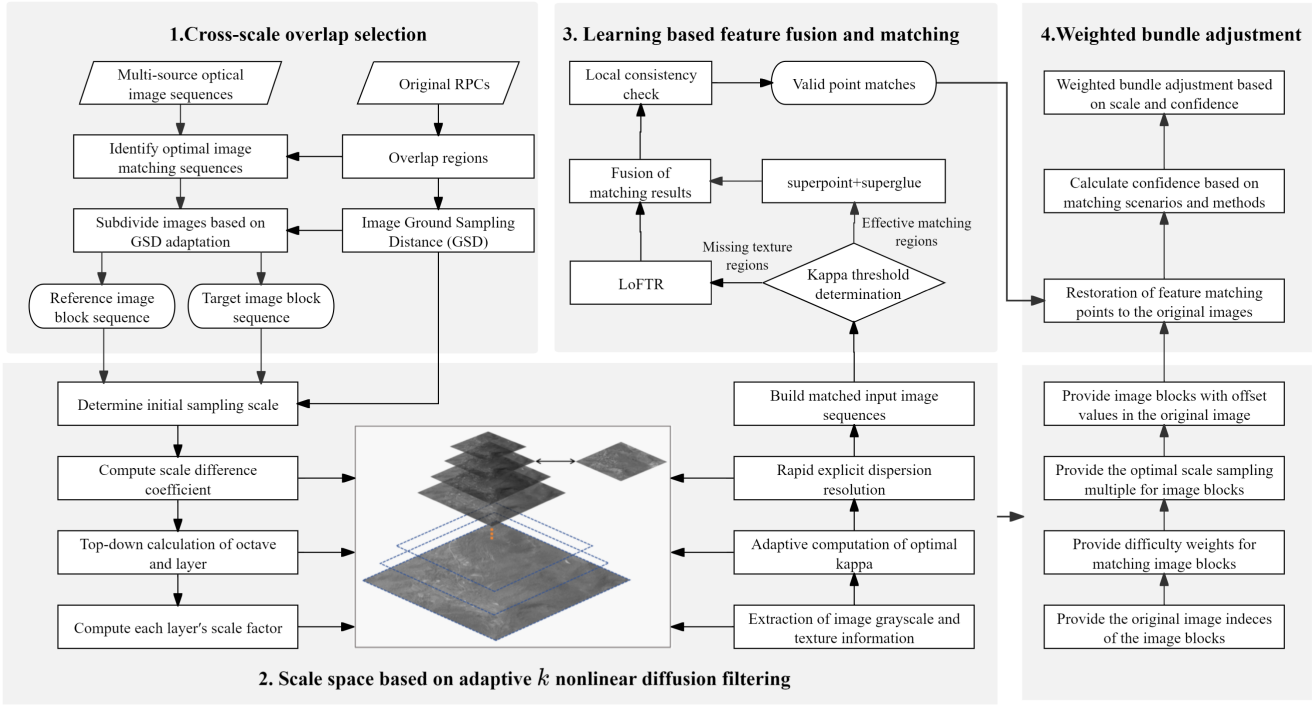

Fig. 2. Illustration of the workflow of the proposed feature matching and image block adjustment method.

where H represents the satellite altitude (vertical distance from the sensor to the ground), f denotes the focal length of the sensor, and p corresponds to the pixel size of the sensor.

In practical applications, satellite images often only provide the corresponding RPC model, and the GSD can be indirectly obtained through the RPC model. The RPC model uses rational functions to associate ground coordinates with image pixel coordinates, with its basic form is

$$\begin{cases} c = \dfrac{P_1(\phi, \lambda, h)}{P_2(\phi, \lambda, h)} \\ r = \dfrac{P_3(\phi, \lambda, h)}{P_4(\phi, \lambda, h)} \end{cases} \tag{2}$$

where P_i is the numerator and denominator polynomial functions of the RPC model; c and r are the column and row pixel coordinates in the image; and ϕ, λ, and h are the latitude, longitude, and elevation of the ground point.

To clarify the specific form of P_i, each polynomial P_i is defined as a third-order rational function with cross terms, subject to the constraint that the sum of exponents does not exceed 3 (i.e., maximum degree of 3). The explicit expression is

$$P_i(\lambda, \phi, h) = \sum_{m=0}^{3} \sum_{n=0}^{3} \sum_{k=0}^{3} a_{\text{mnk}}^{(i)} \cdot \lambda^m \cdot \phi^n \cdot h^k \tag{3}$$

where $a_{\text{mnk}}^{(i)}$ denotes the polynomial coefficient for P_i, which is provided by satellite manufacturers (typically included in RPC parameter files); m, n, and k are nonnegative integer exponents corresponding to the longitude (λ), latitude (ϕ), and elevation (h) terms, respectively.

The GSD is estimated by inversely projecting the coordinates of the image center pixels (c_0, r_0) to the coordinates of the ground (λ_0, ϕ_0, h_0) using the RPC model. Subsequently, moving one pixel horizontally or vertically to $(c_0 \pm 1, r_0)$ or $(c_0, r_0 \pm 1)$, the new ground coordinates (λ', ϕ', h') are obtained via RPC inversion. The ground Haversine distance can be calculated through [47]

$$\text{GSD} = D_H\left((\lambda_0, \phi_0), (\lambda', \phi')\right) \tag{4}$$

where D_H means the great-circle distance between two geographic coordinates.

The image with the minimum GSD (highest spatial resolution) is chosen as the reference image. Then, the four corner pixel coordinates $(c_{\min}, r_{\min})$, $(c_{\max}, r_{\min})$, $(c_{\min}, r_{\max})$, and $(c_{\max}, r_{\max})$ of the reference image are converted to geographic coordinates (λ, ϕ, h) using RPC forward projection. These coordinates are then projected onto the target image's pixel coordinate system to obtain the projected bounding box $[c'_{\min}, c'_{\max}] \times [r'_{\min}, r'_{\max}]$. The initial overlapping area is determined by intersecting this bounding box with the target image's coverage area

$$\begin{aligned} c_{\text{overlap-min}} &= \max\left(c'_{\min}, c_{\min}\right) \\ c_{\text{overlap-max}} &= \min\left(c'_{\max}, c_{\max}\right) \\ r_{\text{overlap-min}} &= \max\left(r'_{\min}, r_{\min}\right) \\ r_{\text{overlap-max}} &= \min\left(r'_{\max}, r_{\max}\right). \end{aligned} \tag{5}$$

To refine the overlapping region, a dense sampling strategy is employed by uniformly sampling points across the reference image (e.g., every n pixels). The projection steps are repeated for these samples, and the resulting points are connected to form a precise polygonal overlapping area.

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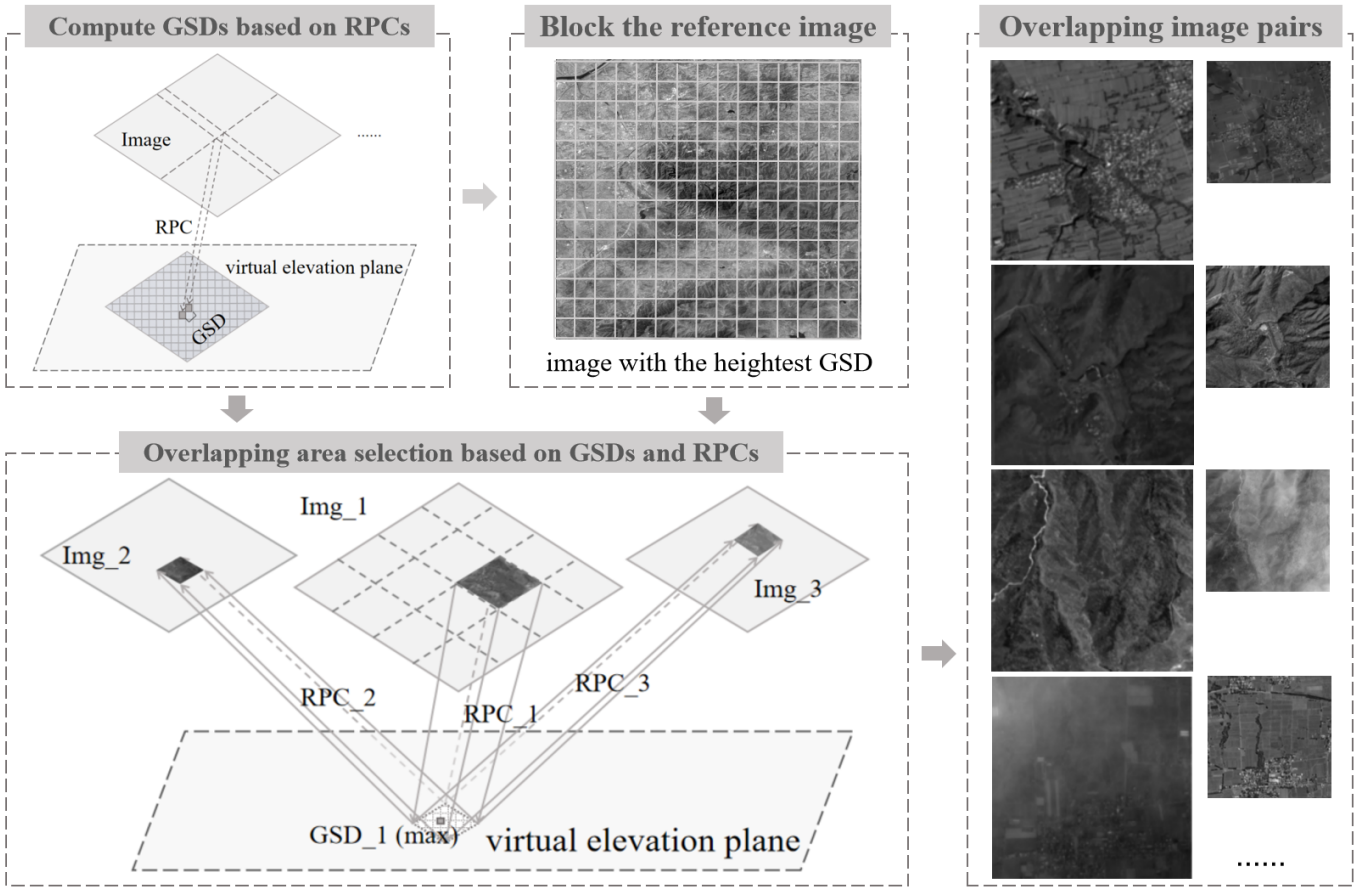

Fig. 3. Schematic of cross-scale overlapping area selection based on RPC and GSD.

B. Scale Space Based on Adaptive κ Nonlinear Diffusion Filtering

1) Adaptive κ Nonlinear Diffusion Filtering: Nonlinear diffusion filtering can effectively preserve the structures while smoothing the texture and removing noise. Thus, we choose nonlinear diffusion filtering to achieve better structures [21]. KAZE uses adaptive κ to balance noise suppression and detail preservation by synergizing gradient magnitude analysis and global texture assessment. The exponential decay diffusion coefficient function is defined as

$$c\left(\|\nabla I\|\right) = \exp\left(-\left(\frac{\|\nabla I\|}{\kappa}\right)^2\right) \tag{6}$$

where $c(\|\nabla I\|)$ is the diffusion coefficient at a given point; $\|\nabla I\|$ is the magnitude of the gradient of the image I at that point, representing the rate of intensity change; and κ is a threshold parameter that modulates the balance between edge preservation and noise smoothing.

To enhance computational efficiency and ensure numerical stability during the diffusion process, we use FED for further optimization. FED utilizes a variable time-step sequence in the iterative solution of the diffusion equation, allowing for larger time steps while maintaining stability. The time steps Δt_k are defined as

$$\Delta t_k = \frac{T}{M}\sin^2\left(\frac{\pi\left(k-0.5\right)}{2K}\right), \quad k = 1, 2, \ldots, K \tag{7}$$

where Δt_k is the time step for the kth iteration; T is the total diffusion time; M is the number of cycles; and K is the total number of iterations (with $K = M \times N$, where N is the number of iterations per cycle), and k is the iteration index. This strategy ensures numerical stability while permitting larger time steps for accelerated convergence.

Since the scene of a satellite image is complex, including a variety of objects and textures, the gradient of images affected by noise may not accurately capture the structural features. To address this challenge, we incorporate the global variance V_{global} into the computation of κ to account for the overall statistical characteristics of the image. The global variance captures the distribution of pixel intensities across the entire image, providing a measure that reflects the overall image texture and variability. By considering both the gradient and the global variance, the adaptive κ value becomes more robust against the effects of complex scenes and small objects on the gradients. This ensures that invalid noise is more effectively suppressed during filtering while preserving important image details. Thus, the final calculation step of κ can be defined as follows:

$$\begin{cases} \kappa_{\text{gradient}} = \text{median}\left(\|\nabla I\|\right) \\ V_{\text{global}} = \dfrac{1}{MN}\displaystyle\sum_{i,j}\left(I_{i,j} - \mu_{\text{global}}\right)^2 \\ \kappa_{\text{texture}} = K_0 + \gamma V_{\text{global}} \\ \kappa = \lambda_1 \kappa_{\text{gradient}} + \lambda_2 \kappa_{\text{texture}}, \quad \text{where } \lambda_1 + \lambda_2 = 1 \end{cases} \tag{8}$$

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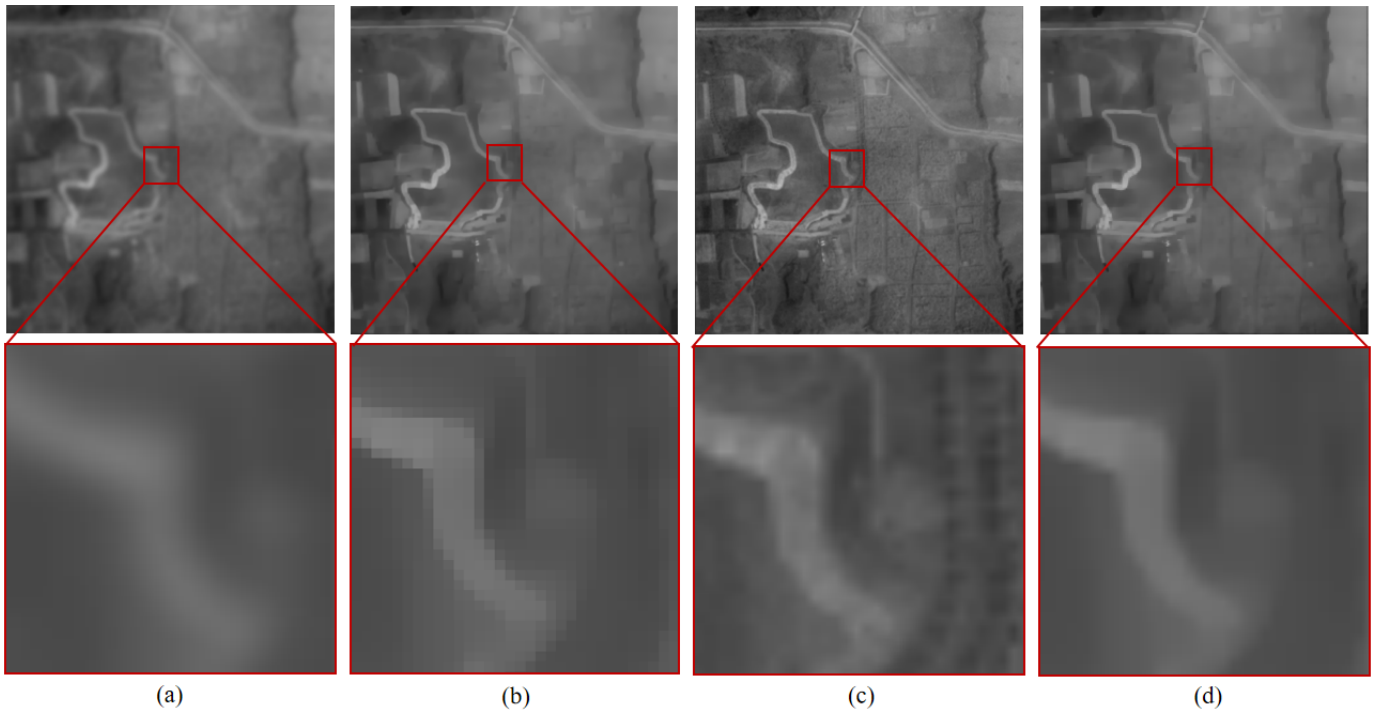

Fig. 4. Comparison of different sampling methods. (a) Gaussian filtering. (b) Simple downsampling. (c) Nonlinear diffusion filtering with KAZE. (d) Nonlinear diffusion filtering with the proposed method.

where κ_{gradient} denotes the median value of the image gradient magnitude $\|\nabla I\|$; V_{global} is the global variance of the image, with $M \times N$ representing the total number of pixels in the image, $I_{i,j}$ the gray value of the pixel at row i and column j, and μ_{global} the global mean gray value of the image; κ_{texture} is the texture-related component of κ, where K_0 is a predefined base value (set to 0.5 empirically) and γ is the weight coefficient of V_{global} (fixed at 0.3 based on prior studies); λ_1 and λ_2 are the weight coefficients of κ_{gradient} and κ_{texture}, respectively, satisfying the constraint $\lambda_1 + \lambda_2 = 1$, and they are finally set to 0.6 and 0.4 via fivefold cross-validation.

It can be seen from Fig. 4(a)–(d), our method outperforms traditional Gaussian filtering. It avoids the aliasing artifacts caused by simple downsampling [see Fig. 4(b)] and produces cleaner results than methods like KAZE, which rely solely on a fixed percentile of the gradient histogram and may retain unwanted noise. The filtering proposed in this article ensures that the unilateral local scale space we construct is robust and precise, enabling accurate feature detection across diverse scales in complex remote sensing imagery.

2) Construction of Local Continuous Scale Space: To mitigate scale uncertainties arising from RPC model errors and terrain undulations in cross-scale remote sensing image matching, this study proposes a local continuous scale space construction framework guided by the relative GSD between reference and target images. The methodology unfolds as follows:

The relative scale difference S, quantifying the spatial resolution ratio between images, is initially derived as

$$\begin{cases} S = \dfrac{\text{GSD}_{\text{tgt}}}{\text{GSD}_{\text{ref}}} \\ \sigma_{\text{tgt}} = \sigma_0 \times S \end{cases} \tag{9}$$

where GSD_{ref} and GSD_{tgt} represent the ground resolutions of the reference and target images, respectively, and $\sigma_0 = 1.6$ is the SIFT-inspired baseline scale.

To account for scale perturbations, a deviation range $[\sigma_{\min}, \sigma_{\max}]$ is established around σ_{tgt}, controlled by a dynamically adjusted coefficient δ

$$\begin{cases} \delta = \delta_0 + (S - 1) + \theta \cdot \dfrac{U}{U_{\max}} \\ \sigma_{\min} = \sigma_{\text{tgt}} (1 - \delta) \\ \sigma_{\max} = \sigma_{\text{tgt}} (1 + \delta) \end{cases} \tag{10}$$

where δ_0 is a predefined offset (set to 0.05 empirically to ensure minimum scale flexibility); U is the terrain undulation metric (e.g., elevation standard deviation); $U_{\max}$ is the maximum terrain undulation of the study area (normalizing U to $[0, 1]$ to avoid extreme terrain interference); and θ is the dedicated regulatory parameter for U (controlling U's influence without overriding S), set to 0.2 via fivefold cross-validation [(maximizing matching success rate (SR)].

The scale space hierarchy is subsequently structured following the standard SIFT methodology [7]. Specifically, the scale range $[\sigma_{\min}, \sigma_{\max}]$ is subdivided into octaves and layers using the octave law and geometric progression factor to ensure continuous scale transitions, as per the original SIFT algorithm.

Gaussian smoothing is applied at each scale to construct the scale space representation

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y) \tag{11}$$

where $G(x, y, \sigma)$ denotes the Gaussian kernel, and $I(x, y)$ is the input image. This process effectively suppresses distortions induced by RPCs while preserving image features critical for accurate matching.

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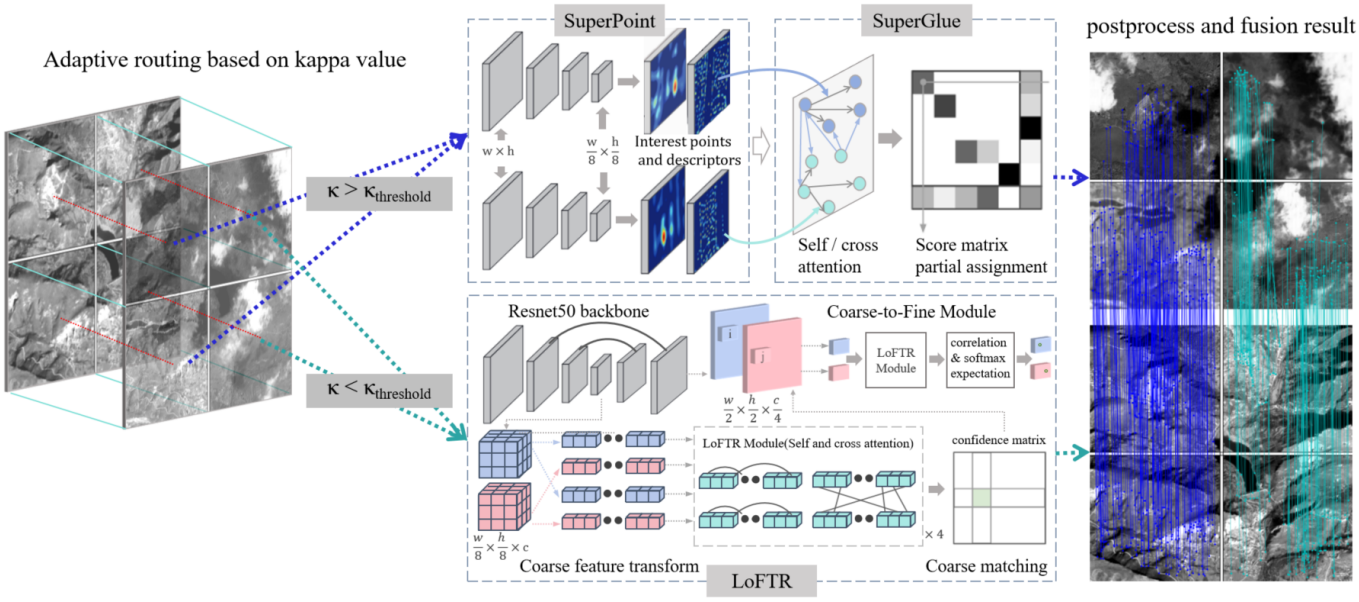

Fig. 5. Schematic of the fusion matching workflow with adaptive routing for multiscale satellite images.

By aligning the scales of the reference and target images in this manner, we enhance the robustness and precision of feature detection across diverse imaging conditions. This approach allows for effective handling of images with different resolutions and varying terrain characteristics, which is crucial for remote sensing applications.

C. Learning-Based Feature Fusion and Matching

Recently, deep learning methods have demonstrated superior performance. This section presents a high-precision deep learning matching strategy based on the analysis of parameter κ introduced in Section III-C. By quantifying local texture intensity with κ (higher values indicate richer textures, while lower values imply weaker textures), our method employs a dynamic routing mechanism to adaptively switch between feature matching modules. SP + SG is activated for high-texture regions to leverage its local feature detection accuracy, while LoFTR is invoked for low-texture regions to utilize its global context-awareness. This hybrid framework synergizes the strengths of both methods through adaptive routing decisions, achieving remarkable improvements in satellite image matching across diverse texture conditions.

A threshold $\kappa_{\text{threshold}}$ is set to distinguish between richly textured regions and weakly textured regions. To ensure the threshold is adaptive to the texture characteristics of the input image, $\kappa_{\text{threshold}}$ is calculated based on the statistical distribution of κ values across the entire image—first computing the mean of all local κ values, then multiplying by an adaptive coefficient within the range $[0, 1]$. The integrated formula is defined as follows:

$$\begin{cases} \mu_\kappa = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} \kappa_i \\ \kappa_{\text{threshold}} = \mu_\kappa \times \omega \end{cases} \tag{12}$$

where μ_κ denotes the mean value of κ computed for all local regions in the image, N is the total number of local regions in the image, κ_i is the κ value of the ith local region, and ω is an adaptive coefficient with a value range of $[0, 1]$, set to 0.6 via cross-validation [balancing matching SR and bundle adjustment accuracy (BAA)]

$$\text{Matching Method} = \begin{cases} \text{SP+SG}, & \text{if } \kappa \geq \kappa_{\text{th}} \\ \text{LoFTR}, & \text{if } \kappa < \kappa_{\text{th}}. \end{cases} \tag{13}$$

As illustrated in Fig. 5, the fusion matching workflow proceeds as follows: First, the input image pair (reference and target images) is partitioned into multiple local blocks, based on the overlapping region identified in Section III-A. For each block, κ is computed to assess texture richness, guiding adaptive routing: blocks with $\kappa \geq \kappa_{\text{threshold}}$ (rich texture) use the CNN-based SP + SG framework to generate matched point pairs, while blocks with $\kappa \leq \kappa_{\text{threshold}}$ (weak texture) rely on the transformer-based LoFTR.

The two matching modules operate via distinct yet complementary mechanisms. The SP + SG Framework functions in two sequential stages: the SuperPoint (CNN Backbone) adopts a visual geometry group (VGG)-style encoder as its backbone for keypoint detection and descriptor generation, extracting hierarchical image features through stacked convolutional layers, filtering reliable keypoints via confidence screening and nonmaximum suppression, and generating 64-D descriptors for each keypoint; the Superglue takes the outputs of SuperPoint as input and employs a GNN to model interimage feature correlations, where the GNN refines feature representations by capturing contextual dependencies between keypoints, enabling accurate prediction of matching relationships to yield initial point pairs.

The Transformer-based LoFTR leverages a Transformer architecture with a ResNet50 backbone for feature extraction, enhancing feature discriminability through two attention mech-

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anisms: self-attention refines intraimage features by modeling dependencies between local regions of a single image, while cross-attention establishes interimage feature associations to align corresponding regions across the reference and target images. After feature enhancement, LoFTR uses a two-stage matching pipeline (coarse matching for initial correspondence localization and fine matching for subpixel precision) to generate initial matched pairs.

Specific details regarding network layers (e.g., number of convolutional/transformer layers) and feature dimension transformations are annotated in Fig. 5. Post-processing of all initial matched pairs (from both modules across blocks) is performed to derive globally consistent point pairs. For local outlier removal, each block uses the random sample consensus (RANSAC) algorithm (coupled with a fundamental matrix model) to filter out outlier pairs with large reprojection errors, retaining geometrically consistent inliers. For global consistency verification, all local inliers are integrated, and a scene-adaptive geometric model (homography matrix for planar scenes or fundamental matrix for nonplanar scenes) is applied to eliminate global outliers that violate overall geometric constraints; the remaining inliers constitute the final globally matched point pairs, which are used for subsequent BBA.

D. Weighted Adjustment Based on Scale and Confidence

In the joint geometric positioning of multiscale optical images, bundle adjustment, as the core technology, plays a vital role. Given the remarkable differences in the ground coverage areas corresponding to image points of images with different resolutions, when constructing the collinearity equation, the additional variables introduced by the scale effect must be fully taken into account. This undoubtedly significantly increases the complexity of constructing the error equation and the process of parameter solving [48].

To obtain the observation equation of the combined adjustment model, the rational function model (RFM) is subjected to a weighted Taylor expansion [49], yielding

$$\mathbf{V} = \mathbf{A}\mathbf{X}_1 + \mathbf{B}\mathbf{X}_2 - \mathbf{L}, \mathbf{P}. \tag{14}$$

This equation establishes a relationship between ground points and image tie-points. Specifically, $\mathbf{X}_1$ denotes the correction vector for the bias-compensation coefficients, $\mathbf{X}_2$ represents the correction vector for the ground coordinates of the tie-points, $\mathbf{V}$ is the residual vector, $\mathbf{L}$ corresponds to the observed values of the tie-points, $\mathbf{P}$ is the weight matrix of the tie-points, and $\mathbf{A}$ and $\mathbf{B}$ are the partial derivatives with respect to the bias-compensation coefficients and the ground coordinate values of the tie-points, respectively. When a specific image is selected as the reference image, the coefficient matrix can be further derived based on the compensation model.

Regarding the determination of weights, different strategies are employed based on the availability of high-precision stereo image pairs. When at least one high-precision stereo pair is available, forward intersection is performed first to obtain ground points from both high- and low-precision stereo pairs. These ground points are then back-projected into the image space, and the residuals R_{rc} of the tie-points in the image space are calculated. Here, R_{rc} represents the residuals of the tie points in the image space relative to their theoretical positions in the image-plane coordinate system.

The weights of the tie-points in the image space are determined as

$$P_i = \frac{1}{R_{\mathrm{rc}} \times \mathrm{GSD}} \times C \tag{15}$$

where C is the confidence adjustment coefficient, and its value is determined by comprehensively considering factors such as the texture richness of the image block and the feature-matching method. The texture richness of the image block can be measured based on the previously calculated κ value. The larger the κ value, the richer the texture of the image block, and the higher the confidence of the corresponding homologous points; conversely, the lower it is. The κ value is normalized to obtain a quantified value *kappa* with a value range of $[0, 1]$ to represent the texture richness. In terms of the feature-matching method, for example, the points obtained by using SP + SG have a relatively high confidence, while the confidence of LoFTR is relatively low. Accordingly, a quantified value M for the feature-matching method is set, with SP + SG taking a value of 1, LoFTR taking a value of 0, and other methods taking values between 0 and 1 according to the confidence level. By incorporating the image resolution and confidence into the weight calculation, not only is the accurate geometric information provided by high-resolution images emphasized, but also the reliability of homologous points is taken into account.

IV. Experiments

A. Experimental Data, Metrics, and Configuration

1) Experimental Area 1: We selected stereo image pairs from GaoFen-7 (GF-7) and ZiYuan-3 (ZY-3) satellites as experimental data for Earth observation, with a maximum resolution difference of up to 5.38 times. The test area is located between latitudes 33.89°N and 34.96°N, and longitudes 112.45°E and 113.68°E, covering the Songshan region in Henan, China. The elevation ranges from 29 to 1492 m. This area includes 23 pairs of GF-7 stereo images, each comprising a forward-looking and a backward-looking image with resolutions of 0.8 and 0.65 m, respectively, collected between May 2020 and April 2021. In addition, there are four pairs of ZY-3-01 satellite stereo images, each consisting of forward, backward, and nadir views, with resolutions of 3.5 m for forward and backward views and 2.1 m for nadir views, collected from November 2015 to October 2016. An overview of the images is illustrated on the left side of Fig. 6.

2) Experimental Area 2: Mars context imager (CTX) and high-resolution imaging science experiment (HiRISE) images were selected as experimental data for a more challenging scenario, with a resolution difference of up to 20 times between CTX and HiRISE. The Oxia Planum region on Mars was chosen as the test area; it is a significant geological area near the Martian equator and was the selected landing site for the ExoMars mission [50]. Specifically, it is located between latitudes 16°N and 24°N, and longitudes 335°E and 345°E on Mars. This region is rich in clay minerals, indicating the

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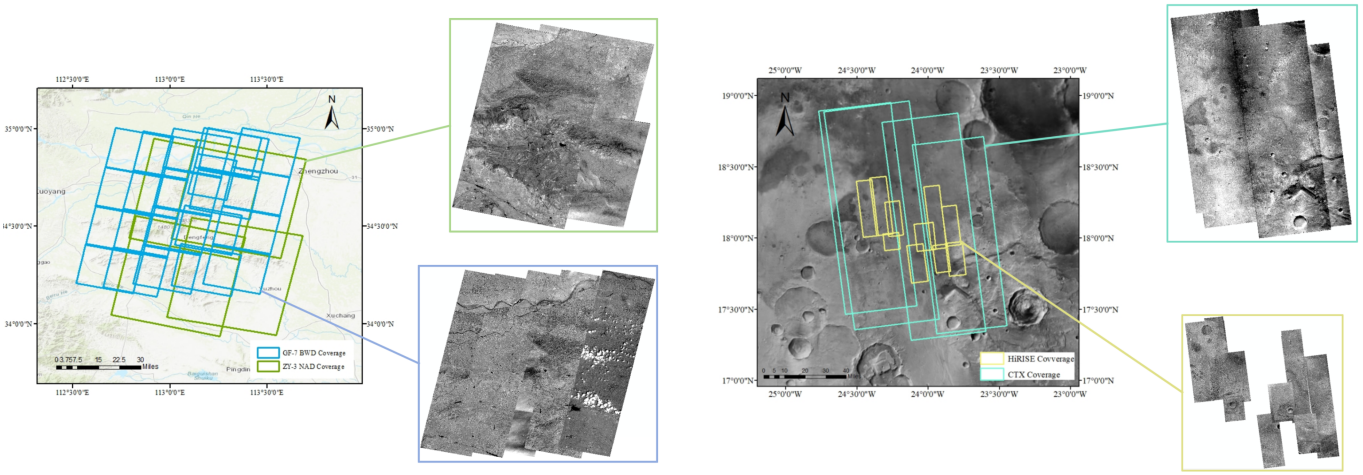

Fig. 6. Overview of the satellite images used in the experiments. (Left) GF-7 and ZY-3 images of experimental area 1. (Right) CTX and HiRISE images of experimental area 2.

past presence of liquid water, and has an elevation range from approximately −2500 to −2000 m. The images were collected between 2008 and 2016. An overview of the images is illustrated on the right side of Fig. 6.

In our experiments, we evaluated the proposed method using several metrics: RT per block, matching success rate (SR), average number of matching (ANM), average inlier ratio (AIR) per block, global inlier ratio (GIR), epipolar error (EE), and BAA. RT measures the computational efficiency of the algorithm. SR represents the proportion of image pairs successfully matched (with more than 15 matching points). ANM indicates the total number of feature matches obtained before outlier removal. AIR is the average ratio of inlier matches retained after applying the RANSAC algorithm per block. GIR represents the ratio of inlier matches to the total number of matches across the entire image pair after a global RANSAC process. EE measures how well the matched point pairs satisfy the epipolar geometric constraint. BAA evaluates the satellite images' alignment consistency (unit: reprojection residual in pixels) after bundle adjustment.

To align with the CNN and Transformer architectures detailed in Section III-C and ensure the method's reliability, we specify key experimental implementation details:

The CNN input sample size (image block size) for inference is 400×400 pixels (high-resolution images, e.g., GF-7, HiRISE) and 2000×2000 pixels (low-resolution images, e.g., ZY-3, CTX), matching the requirements of the VGG-style encoder (fixed-size input for consistent feature mapping) and ResNet50 backbone (adaptable to large inputs via residual downsampling). The SuperPoint encoder uses strided convolution (total downsampling rate of 8) without pooling layers to preserve texture details, while ResNet50 adopts max-pooling (2×2 kernel, stride of 2) after the first convolutional layer to reduce spatial dimensions.

Model training uses a dual-source dataset: SuperPoint is pretrained on COCO 2017 (80k natural scene images) and fine-tuned on 8316 satellite image blocks (WHU MVS/Stereo Dataset, covering mountainous, flat, and weak-texture scenarios); LoFTR is pretrained on MegaDepth (10k depth-annotated images) and fine-tuned on the same satellite blocks. Key hyperparameters are as follows: SuperPoint uses batch size = 16, Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$), learning rate = 10^{-4} (halved every 5 epochs), 20 training epochs, and loss function of binary cross-entropy + triplet loss (margin = 0.2); LoFTR uses batch size = 8, AdamW optimizer (weight decay = 10^{-4}), learning rate = 5×10^{-5} (halved every 5 epochs), 20 training epochs, and loss function of cross-entropy $+L1$ loss.

Inference is conducted on an Intel i7-14700K CPU + NVIDIA RTX 4080 GPU workstation, with image partitioning processed in parallel (ten threads) for large-scale data acceleration. For fair comparison with competing methods, unified parameters include max keypoints = 5000 (SIFT, KAZE, and SuperPoint), NMS radius = 4 pixels, RANSAC confidence = 99% (fundamental matrix for outlier removal), and keypoint detection thresholds (0.04 for SIFT, 0.001 for KAZE, 0.8 for LoFTR).

B. *Effectiveness of Optimal Scale (OS)*

To evaluate the effectiveness of the proposed local unilateral nonlinear diffusion filtering in multiscale satellite image matching, we conducted experiments on two datasets: Earth observation image pairs and Mars observations. We compared four methods: Simple downsampling, where the target image is downsampled to match the size of the reference image; SIFT's Gaussian scale space, constructing image pyramids by sequentially downsampling both images at a factor of 1/2; AKAZE's nonlinear scale space, using a similar approach but utilizing AKAZE's nonlinear diffusion filtering; and our proposed adaptive local nonlinear scale space, which constructs a local scale space using unilateral nonlinear diffusion filtering based on the κ value.

1) Earth Observation Results: The first dataset was selected from Earth observation scenes. It consists of two pairs of images captured by the aft-looking camera of the GF-7 satellite and the nadir-looking camera of the ZY-3 satellite, exhibiting approximately a fivefold scale difference. Fig. 7 shows the Earth observation images of the Songshan region and the challenging scenarios encountered in these experiments.

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TABLE I

COMPARISON OF SCALE SPACE CONSTRUCTION METHODS ON EARTH OBSERVATION IMAGES (GF-7 AND ZY-3)

Method	RT (s)	ANM	AIR (%)	EE (pixels)
Simple down-sampling	0.08	425.21	85.54	2.86
SIFT's scale space	0.75	215.65	87.17	2.10
AKAZE's scale space	0.86	253.82	91.34	1.52
Proposed Method	0.15	467.19	92.25	1.32

The selected images contain challenging areas, including significant radiometric differences, object discrepancies caused by different acquisition times, viewpoint variations, and cloud or fog occlusions.

For each method, the high-resolution images were processed to match the scale of the low-resolution images. The input matching pairs were all obtained using the cross-scale overlapping area selection method proposed in Section III; these matching image block pairs ensure sufficient overlapping areas despite different scale differences.

A total of 792 pairs of image blocks were obtained for subsequent feature matching and evaluation. All image block pairs were subjected to feature matching using SP + SG, and four evaluation metrics—RT, ANM, AIR, and EE—were used to assess the effectiveness of different scale space construction methods.

In this context, RT represents the time taken to construct the scale space, ANM and AIR are obtained from the layer with the highest number of matched points in the scale space, and EE is computed based on the original size of the GF-7 image blocks.

From Table I, it is evident that the proposed algorithm exhibits outstanding performance across multiple metrics.

Regarding the commonly used Gaussian filtering method (as implemented in SIFT's scale space), the unilateral local nonlinear scale space with an adaptive κ value effectively preserves edge contours, significantly increasing the number of initial matches (ANM of 467.19 compared with 215.65 for Gaussian filtering). Moreover, it enhances the accuracy of restoring downsampled feature point coordinates to the original image size due to the more distinct features preserved.

The improved clarity of features stems from better preservation of edge information, which enhances matching accuracy. This is reflected in the higher AIR per block (AIR of 92.25%) and lower EE (EE of 1.32 pixels) compared with Gaussian filtering (AIR of 87.17%, EE of 2.10 pixels).

In comparison to methods that directly employ simple downsampling, the proposed approach significantly improves the accuracy of feature points by effectively avoiding aliasing effects. Simple downsampling, while computationally efficient (RT of 0.08 s), suffers from a higher EE (2.86 pixels) and lower AIR (85.54%), indicating less precise matching due to the retention of high-frequency noise and loss of crucial image details.

When compared with the nonlinear filtering method mentioned in AKAZE, the algorithm greatly enhances computational efficiency. AKAZE's nonlinear scale space method, while achieving good matching accuracy (EE of 1.52 pixels, AIR of 91.34%), is time-consuming (RT of 0.86 s) due to the iterative nature of solving complex nonlinear partial differential equations across multiple scales.

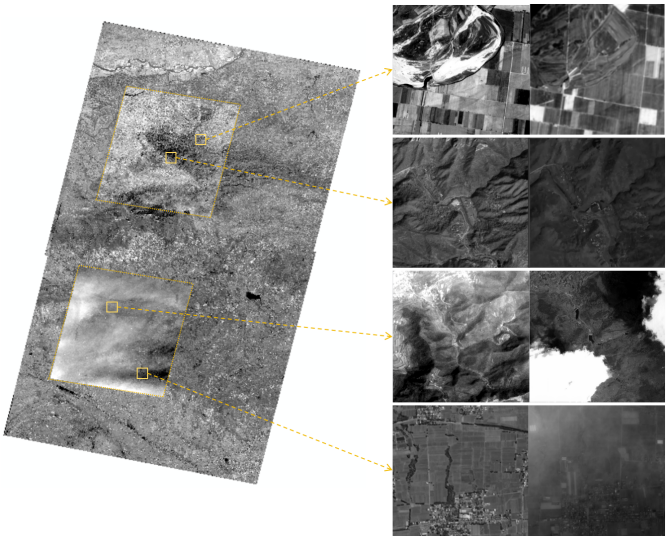

Fig. 7. Earth observation of the Songshan region and difficult scenarios.

TABLE II

COMPARISON OF METHODS ON MARS OBSERVATION IMAGES (HIRISE AND CTX)

Method	RT (s)	ANM	AIR (%)	EE (pixels)
Simple down-sampling	0.45	97.34	83.43	4.75
SIFT's scale space	2.63	87.83	85.66	2.73
AKAZE's scale space	2.89	89.75	87.74	2.52
Proposed Method	1.12	113.54	91.36	1.77

By constructing a unilateral local nonlinear scale space, the proposed method not only improves efficiency (RT of 0.15 s) but also captures scales closer to the optimal through finer scale space intervals. This contributes to higher matching precision, as evidenced by the highest ANM (467.19) and AIR (92.25%) among the compared methods.

As illustrated in Fig. 8, the scale spaces generated by SIFT and AKAZE fail to attain the OS, typically yielding only fixed scale ratios. In contrast, the local scale space constructed by the method proposed in this article is capable of identifying the OS at a finer scale granularity, and the OS varies across different terrains. It can be observed that at the OS, a greater number of points can be obtained.

2) Mars Observation Results: The second dataset is derived from Mars observation scenarios and comprises two pairs of images acquired by the HiRISE and CTX cameras, exhibiting an approximate 20-fold scale disparity.

Excerpts shown in Fig. 9 illustrate the challenging conditions inherent in deep-space imagery compared with terrestrial images, including illumination differences, weak textures, image noise, and significantly larger scale variations.

In these examples, the image blocks using CTX as the reference image all have dimensions of 240 $\times$ 240 pixels, while the corresponding HiRISE image blocks have widths and heights ranging between 5000 and 6000 pixels.

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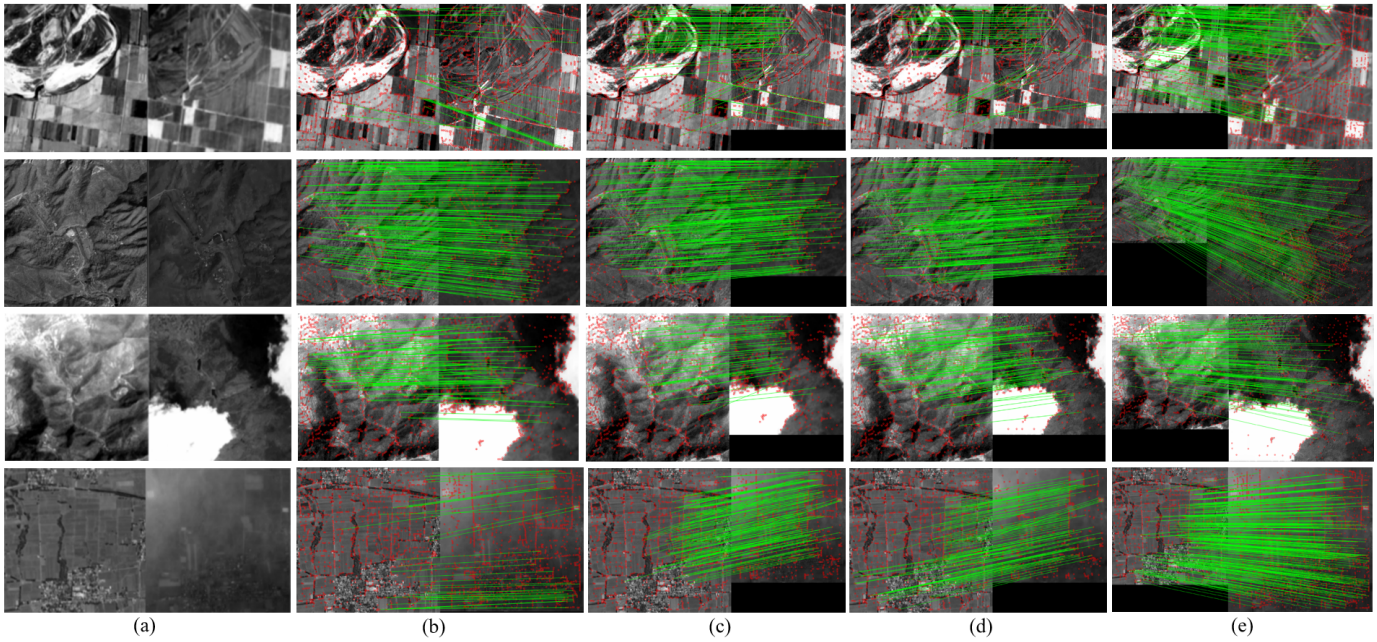

Fig. 8. Feature matching results using Earth observation images with different scale spatial methods. (a) Image pair. (b) Simple downsampling. (c) SIFT's scale space. (d) AKAZE's scale space. (e) Proposed method.

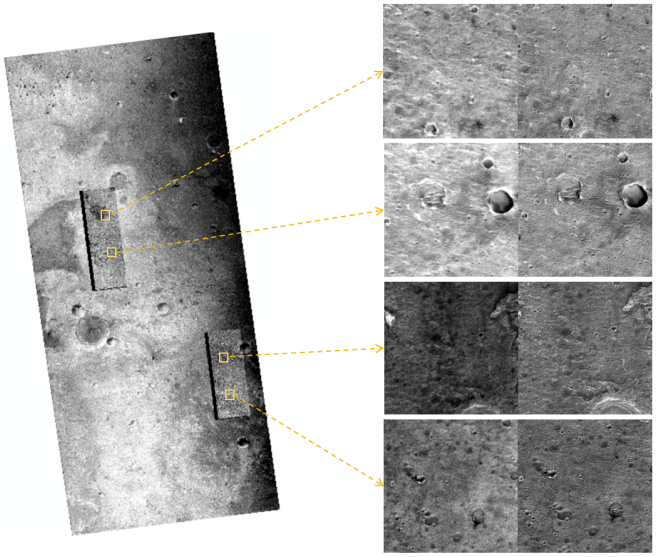

Fig. 9. Mars observation of the Oxia Planum region and difficult scenarios.

The results for the Mars observation image pairs are presented in Table II.

Analyzing the data presented in Table II, we observe that the proposed algorithm continues to outperform the other methods in the context of Mars observation images captured by HiRISE and CTX cameras. However, the improvements are less pronounced compared with the Earth observation results discussed earlier.

The primary reason for this difference lies in the significantly larger scale disparity between the HiRISE and CTX images, which is approximately 20-fold—much greater than the fivefold difference in the Earth observation dataset. This significant resolution difference creates a more challenging scenario for feature matching algorithms.

Despite these challenges, the proposed unilateral local nonlinear diffusion filtering method still offers advantages. The proposed method achieves the highest ANM at 113.54, surpassing both SIFT's scale space (87.83) and AKAZE's nonlinear scale space (89.75). This indicates that the algorithm is able to detect and match more features even under extreme scale differences. It also attains the highest AIR of 91.36% and the lowest EE of 1.77 pixels among all methods tested. This demonstrates superior matching accuracy, which is crucial for reliable image registration and subsequent analysis. Furthermore, with an RT of 1.12 s, the proposed method is considerably faster than both SIFT's (2.63 s) and AKAZE's (2.89 s) scale space construction, making it more practical for processing large datasets.

However, the overall performance metrics are lower than those observed in the Earth observation results, and the matching results are shown in Fig. 10. For instance, the ANM and AIR values are slightly reduced, and the EE is higher compared with the Earth dataset. The greater scale difference in Mars images means that even with nonlinear diffusion filtering, some high-frequency details critical for feature detection may be lost when downsampling from very high resolutions. This affects the distinctiveness of features and the ability to match them accurately. In addition, challenges with weak textures and noise in the Mars surface images impact the performance. Mars images often exhibit weaker textures and higher noise levels compared with Earth imagery. These factors hinder feature detection and matching, as there are fewer prominent features to extract and more noise that can be mistaken for genuine features. Moreover, the impact of extreme scale differences exacerbates the difficulties in aligning features across images. While the proposed method's

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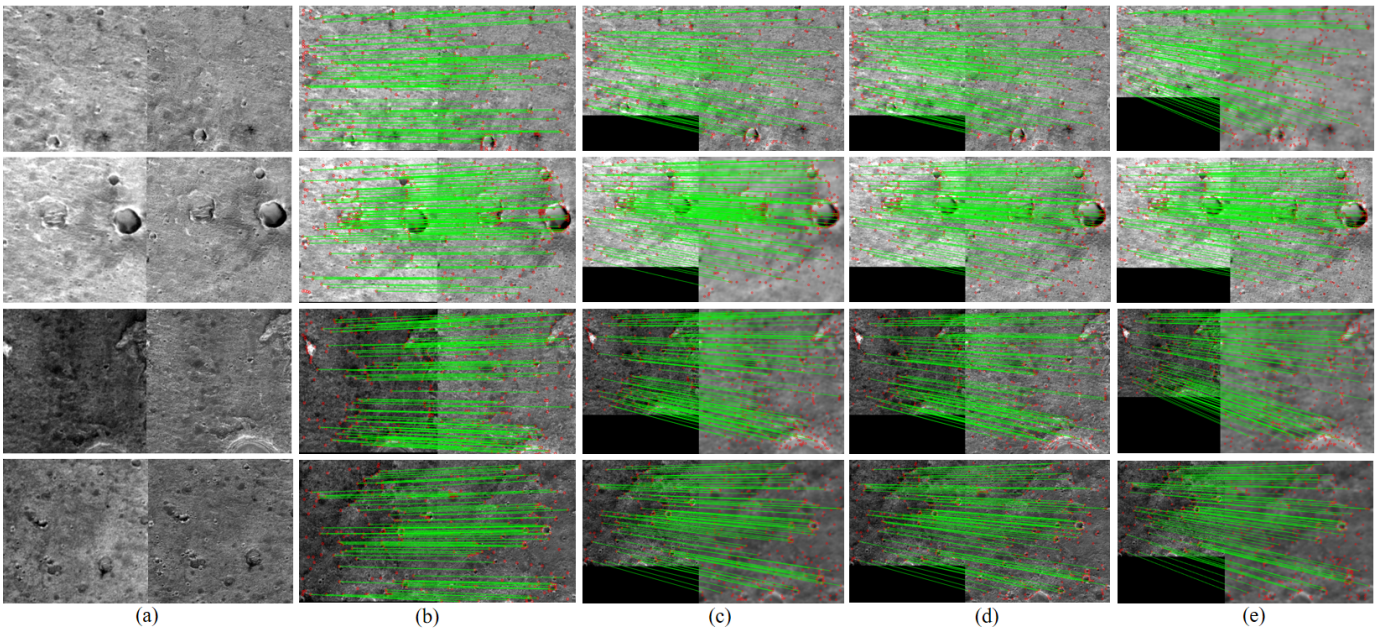

Fig. 10. Feature matching using Mars observation images with different scale spatial methods. (a) Image pair. (b) Simple downsampling. (c) SIFT's scale space. (d) AKAZE's scale space. (e) Proposed method.

ability to find the OS remains beneficial, the matching process inherently becomes less effective as the scale difference increases.

C. Experiment of Multiscale Satellite Images Matching

By integrating an adaptive scale-consistency processing module into the traditional satellite image feature-matching framework, we have developed a system that adaptively processes satellite images with significant scale differences within a unified framework. This approach renders the feature-matching methods—whether feature-based, phase-based, or deep learning-based—agnostic to scale variations, enabling direct comparison of their actual performance.

1) Comparison of Multiple Methods Based on Frameworks: Within our proposed framework, we conducted extensive, large-scale, multiscale feature matching comparisons on Earth observation data. The methods evaluated included traditional feature-based approaches—SIFT [7] and KAZE [21]; phase-based methods—RIFT [27] and CoFSM [29]; and deep learning methods—SP + SG [11], [12], the combination of SuperPoint and LightGlue [13] (hereinafter referred to as SP + LG), LoFTR [14], and Efficient-LoFTR [42](hereinafter referred to as ELoFTR).

For the experiments, 35 image pairs were automatically obtained based on an overlap greater than 20%, focusing solely on heterogeneous matching scenarios. The original images were partitioned using a sampling approach, where blocks were extracted at intervals of 2000 pixels, resulting in 2994 matching blocks. The block sizes were set to 400×400 pixels for the high-resolution images and 2000×2000 pixels for the low-resolution images.

During the image partitioning in our matching framework, elevation data from the SRTM reference DEM with 50-m resolution was utilized. In the scale-consistency processing stage, we applied Gaussian downsampling uniformly to directly reduce the high-resolution image blocks to the same size as the low-resolution blocks. This approach avoids the impact of prescaling on different methods.

Fig. 11 illustrates the distribution of feature points obtained by different methods on large-scale images. Due to the large number of extracted points, not all feature points are displayed; for image blocks where the number of RANSAC-matched points exceeds 20, we select the point with the highest confidence to display. Note that SP + LG and ELoFTR are not separately visualized in the figure, as their feature point distribution patterns are basically consistent with SP + SG and LoFTR, respectively, (no obvious differences in coverage range and density). From visual inspection, we observe that SIFT and KAZE have effective matches only on a small number of image blocks, with KAZE performing slightly better; RIFT and LoFTR perform well, basically covering all effective regions; CoFSM and SP + SG cover over 80% of the areas. This dense and uniformly distributed set of matching points is crucial for accurate alignment in large-scale remote sensing applications.

Table III provides a detailed comparison of the evaluated methods, highlighting metrics such as SR, RT, ANM, AIR, GIR, and BAA. These metrics offer a quantitative assessment of each method's performance within the unified framework, enabling a comprehensive evaluation of their relative effectiveness in large-scale remote sensing applications.

Traditional feature detectors like SIFT and KAZE rely heavily on gradient information and assume brightness constancy and limited appearance changes between images. In the context of multisource remote sensing images, these assumptions are often violated due to significant nonlinear radiometric differences, structural changes over time, and occlusions like severe cloud cover. As a result, SIFT and KAZE exhibit low

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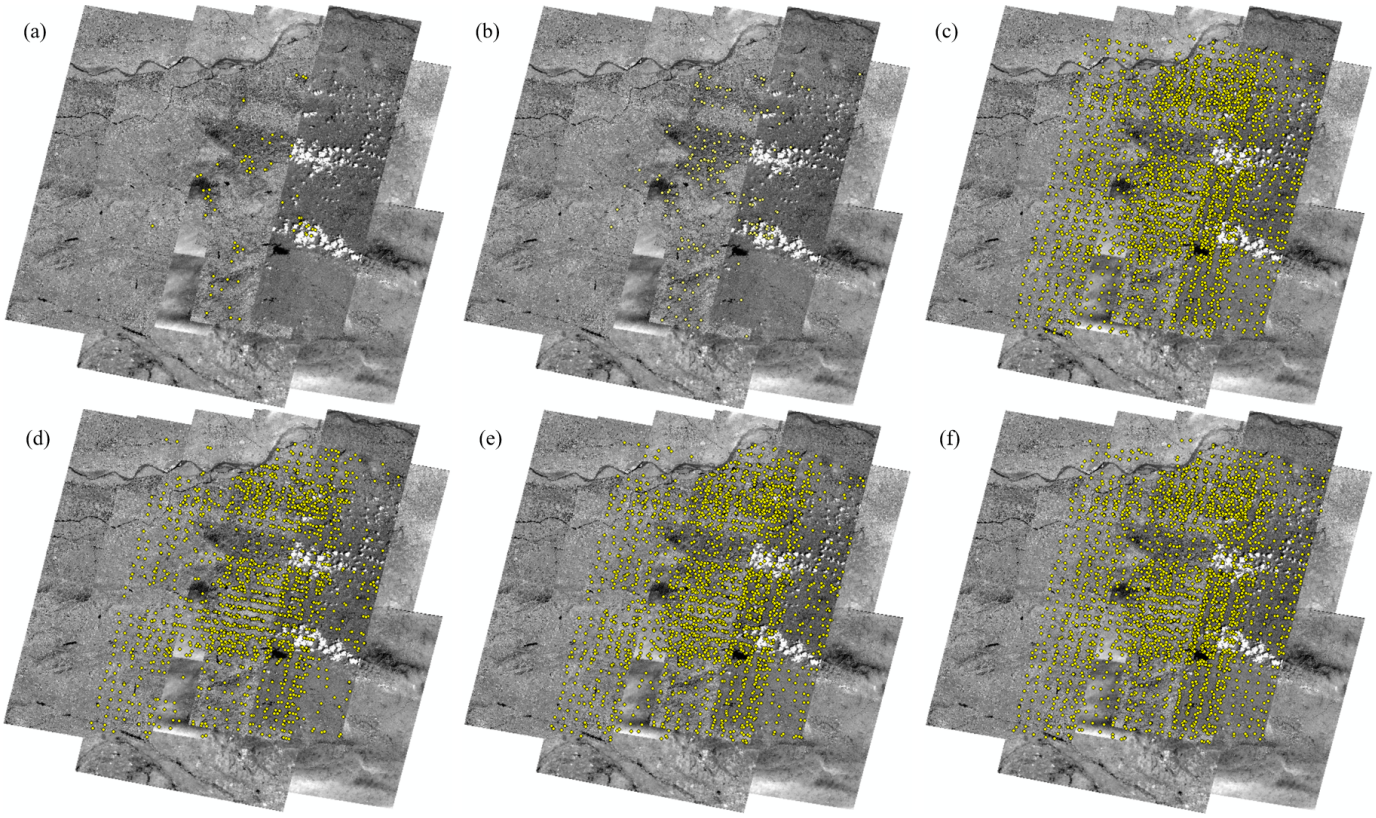

Fig. 11. Feature point distribution effects of different methods on large-scale images. (a) SIFT. (b) KAZE. (c) RIFT. (d) CoFSM. (e) SP + SG. (f) LoFTR.

TABLE III
COMPARISON OF FEATURE MATCHING METHODS ON LARGE-SCALE MULTISOURCE REMOTE SENSING IMAGES

Method	RT (s)	SR (%)	ANM	AIR (%)	GIR (%)	BAA (pixels)	Complexity
SIFT	**0.09**	1.92	76.52	67.48	68.77	–	Keypoint-based ($O(K)$)
KAZE	0.51	4.23	198.55	69.61	62.95	–	Keypoint-based ($O(K)$)
RIFT	31.58	94.70	**1356.24**	64.37	56.79	12.78	Phase-based ($O(N^2)$)
CoFSM	4.73	76.30	384.64	**89.45**	**82.18**	**1.83**	Phase-based ($O(N^2)$)
SP+SG	0.34	83.75	508.96	84.33	78.95	2.16	Keypoint-based ($O(K)$)
SP+LG	0.28	82.90	492.31	83.15	77.62	2.38	Keypoint-based ($O(K)$)
LoFTR	0.31	**95.63**	679.58	82.17	76.56	6.25	Semi-dense ($O(M^2)$, $M = 1/8$ size)
ELoFTR	0.22	94.85	651.74	81.03	75.28	6.52	Semi-dense ($O(M^2)$, $M = 1/32$ size)

SR of 1.92% and 4.23%, respectively. The ANM is also low, with SIFT at 76.52 and KAZE at 198.55. The relatively higher SR of KAZE compared with SIFT reflects, to some extent, the effectiveness of its nonlinear scale space. Due to insufficient valid matches, the BAA cannot be computed (NaN), indicating that these methods fail to provide reliable correspondences for image alignment under challenging conditions.

RIFT demonstrates a high SR of 94.70% and the highest ANM among all methods at 1356.24. This indicates RIFT's capability in detecting a large number of features even under significant geometric transformations. However, its AIR and GIR are relatively low at 64.37% and 56.79%, respectively, suggesting that a substantial proportion of matched features are incorrect. Consequently, the BAA is poor (12.78 pixels), reflecting lower geometric precision. In addition, RIFT has the highest RT (31.58 s), making it less practical for large-scale applications.

CoFSM shows strong performance with an SR of 76.30%, an ANM of 384.64, and notably high AIR and GIR values of 89.45% and 82.18%, respectively. The BAA of 1.83 pixels indicates excellent geometric accuracy after bundle adjustment. CoFSM's robustness can be attributed to its use of coherence filtering for feature detection and a 152-D log-polar descriptor for feature description, which enhances its ability to handle significant geometric transformations and multimodal conditions.

Among deep learning-based methods, SP + SG achieves an SR of 83.75% and an ANM of 508.96, with high AIR (84.33%) and GIR (78.95%) and a BAA of 2.16 pixels, ensuring precise matching and reliable geometric alignment. SP + LG is more efficient than SP + SG, with an RT of 0.28 s (17.6% faster than SP + SG), but its precision metrics are slightly lower—SR at 82.90%, ANM at 492.31, AIR at 83.15%, GIR at 77.62%, and BAA at 2.38 pixels (0.22 pixels higher than SP + SG). LoFTR further improves the SR to 95.63% and the ANM to 679.58, demonstrating strong capability in finding correspondences under challenging conditions. ELoFTR, as an optimized version of LoFTR, has

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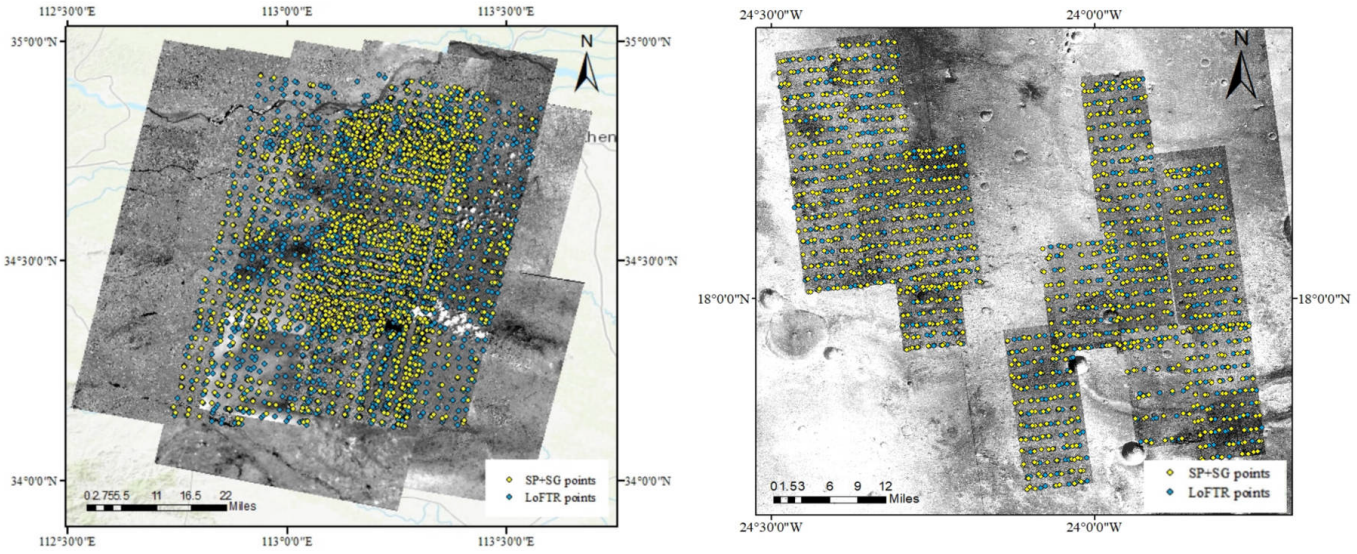

Fig. 12. Feature point distribution results of the proposed method.

TABLE IV
PERFORMANCE COMPARISON OF THE FUSION FEATURE MATCHING METHODS

Methods	RT (s)	SR (%)	ANM	AIR (%)	GIR (%)	BAA (pixels)
SP+SG	0.34	83.75	508.96	**84.33**	78.95	2.16
LoFTR	**0.31**	95.63	**679.58**	82.17	76.56	6.25
Fusion	0.32	95.82	539.20	83.58	79.88	2.08
SP+SG-OS	0.46	85.16	530.45	85.54	79.14	1.91
LoFTR-OS	0.35	**96.59**	700.32	83.79	77.32	5.82
Proposed method	0.42	97.25	612.11	84.27	**81.37**	**1.65**

a shorter RT of 0.22 s (29.0% faster than LoFTR), but its precision is slightly reduced—SR at 94.85%, ANM at 651.74, AIR at 81.03%, GIR at 75.28%, and BAA at 6.52 pixels (0.27 pixels higher than LoFTR). This trade-off between efficiency and precision is the key reason for prioritizing SP + SG and LoFTR in our proposed method: they balance accuracy and robustness more effectively for multiscale satellite image block adjustment.

The outstanding performance of deep learning-based methods is due to their ability to learn robust feature representations from large datasets. They are capable of capturing complex patterns and are more adaptable to radiometric variations, structural changes, and occlusions. This results in higher SR and ANM compared with traditional methods—for instance, SP + SG, SP + LG, LoFTR, and ELoFTR have SR ranging from 82.90% to 95.63% and ANM from 492.31 to 679.58, significantly outperforming SIFT and KAZE.

In terms of computational efficiency, ELoFTR (0.22 s) and SP + LG (0.28 s) are the fastest among deep learning-based methods, while SIFT (0.09 s) is the fastest overall but less robust for multisource remote sensing. Keypoint-based methods (SIFT, KAZE, SP + SG, and SP + LG) have computational cost mainly determined by the number of detected keypoints ($O(K)$, $K \ll N$), whereas phase-based methods (RIFT, CoFSM) require global phase transformations, resulting in higher complexity ($O(N^2)$) and longer RTs. Semi-dense matching methods (LoFTR, ELoFTR) focus attention on downsampled feature maps, with complexity $O(M^2)$ (M is feature map size, e.g., 1/8 or 1/32 of image dimensions). Our proposed method further improves efficiency by adaptively switching modules according to image texture, balancing speed and matching precision.

2) Effectiveness of the Proposed Matching Method: The fusion matching strategy based on κ value proposed in this article integrates the advantages of two learning-based methods. While SP + SG offers high matching accuracy, it often encounters difficulties in extracting effective matching points in many regions. In contrast, LoFTR is efficient with a high SR, capable of improving matching performance in challenging scenarios with only a minimal increase in computation time. This provides more effective matching points and consequently enhances the BAA.

Fig. 12 shows the effectiveness of the proposed feature matching method under the OS. It can be observed that the proposed method significantly improves the quantity and distribution of matching points in both Earth and Mars scenarios, especially in regions where single methods struggle to match. In the figure, the yellow points are obtained by the SP + SG method, and the blue points are obtained by LoFTR when the SP + SG method fails.

Table IV compares the performance of SP + SG, LoFTR, and their fusion method with and without the application of the OS, as proposed in this article. It can be observed that the proposed method shows improvements in SR, ANM, and BAA among other metrics. Notably, the performance is further elevated after applying the OS.

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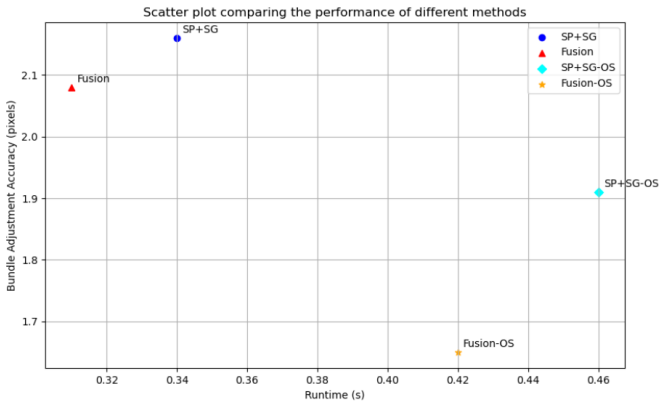

Fig. 13. Scatter plot comparing the performance of different methods.

To visually demonstrate the performance differences among the various methods, Fig. 13 presents a scatter plot with BAA on the vertical axis and RT on the horizontal axis. It is apparent that the proposed method achieves higher matching accuracy while maintaining high efficiency.

Finally, Fig. 14 displays the matching results in local regions, comparing the feature matching performance of different methods in these areas. The visual results of different methods are analyzed as follows. In the SIFT method [see Fig. 14(b)], the number of matching points is few and most of them are incorrect. The RIFT method [see Fig. 14(c)] generates too many matching points, so only 10% of the results are displayed, but there are still many incorrect matches among them. Although CoFSM [see Fig. 14(d)] did not obtain matching results in two of the blocks, it performed very well on the other two blocks where effective matching results were obtained. SP + SG [see Fig. 14(e)] shows poor performance in the two blocks on the right. However, LoFTR-OS is able to obtain effective matches in these two blocks. This fully demonstrates the effectiveness of the proposed method [see Fig. 14(f)].

In summary, the proposed feature matching method achieves superior performance in large-scale remote sensing image matching by integrating the high precision of SP + SG and the high success rate of LoFTR. Particularly, after applying the OS, the proposed method attains a more optimal balance between efficiency and accuracy.

D. Weighted Bundle Adjustment Experiment

Using the tie-points obtained from the feature matching methods described earlier, we performed bundle adjustment. To compare the impact of weighting, we implemented two variants of bundle adjustment. The first is the traditional standard bundle adjustment (SBA), where all tie-points are treated equally without considering their scale or confidence levels. The second is our proposed weighted bundle adjustment (WBA), which assigns weights to tie-points based on the image scale and the confidence coefficients C of the tie-points, as defined in (14).

We evaluated the performance of the SBA and WBA methods by computing the pixel reprojection residuals in the x and y directions separately on images of different resolutions. In addition, we computed the combined residuals in the xy direction, defined as $(x^2 + y^2)^{1/2}$, to assess the overall adjustment accuracy. For the Earth observation data, we assessed the residuals on the high-resolution GF-7 images and the lower resolution ZY-3 images. For the Mars observation data, we evaluated the residuals on the high-resolution HiRISE images and the lower resolution CTX images.

Tables V and VI summarize the adjustment residuals of different methods in the Earth observation dataset (experimental area 1) and the Mars observation dataset (experimental area 2).

The analysis of the results indicates that the WBA method achieves higher accuracy than the SBA across different datasets and resolutions. The improvement is particularly significant in high-resolution images, suggesting that incorporating tie-point confidence and image scale weighting effectively enhances the adjustment accuracy.

In the Earth observation data, for the high-resolution GF-7 images, the combined residual decreased from 1.90 pixels with SBA to 1.60 pixels with WBA, a reduction of approximately 16%. This demonstrates the effectiveness of the weighting strategy in improving adjustment accuracy for high-resolution images. In contrast, for the lower resolution ZY-3 images, the combined residual decreased slightly from 2.93 to 2.85 pixels, showing a smaller improvement.

For the Mars observation data, the high-resolution HiRISE images showed a significant reduction in the combined residual from 3.41 pixels with SBA to 1.57 pixels with WBA, a substantial reduction of about 54%. This indicates that the WBA method effectively improves adjustment accuracy when processing high-resolution images with larger residuals. In comparison, for the lower resolution CTX images, the combined residual decreased slightly from 2.35 to 2.22 pixels, indicating a smaller improvement.

These results suggest that the WBA method, by introducing tie-point confidence and image scale weighting, better balances the influence among different images and feature points. The improvement in adjustment accuracy is more pronounced in high-resolution images. This may be because high-resolution images contain more details and information, and the weighting strategy can more effectively utilize this information to enhance the accuracy of the adjustment results.

V. Conclusion

We have presented a learning-based feature matching framework enhanced by nonlinear diffusion filtering, specifically designed for multiscale optical satellite images. By constructing a local unilateral nonlinear scale space for high-resolution images, employing an adaptive κ-value nonlinear diffusion filter to enhance feature extraction, and adaptively fusing matching strategies between SP + SG and LoFTR based on local texture richness, our method effectively addresses the challenges inherent in multiscale satellite image matching. Experimental results indicate that our approach surpasses traditional and state-of-the-art deep learning methods in terms of accuracy, robustness, and computational efficiency. By effectively handling scale variations, geometric distortions, and radiometric differences, our framework provides a practical solution for multiscale optical satellite image matching, which is of significant importance for applications such as

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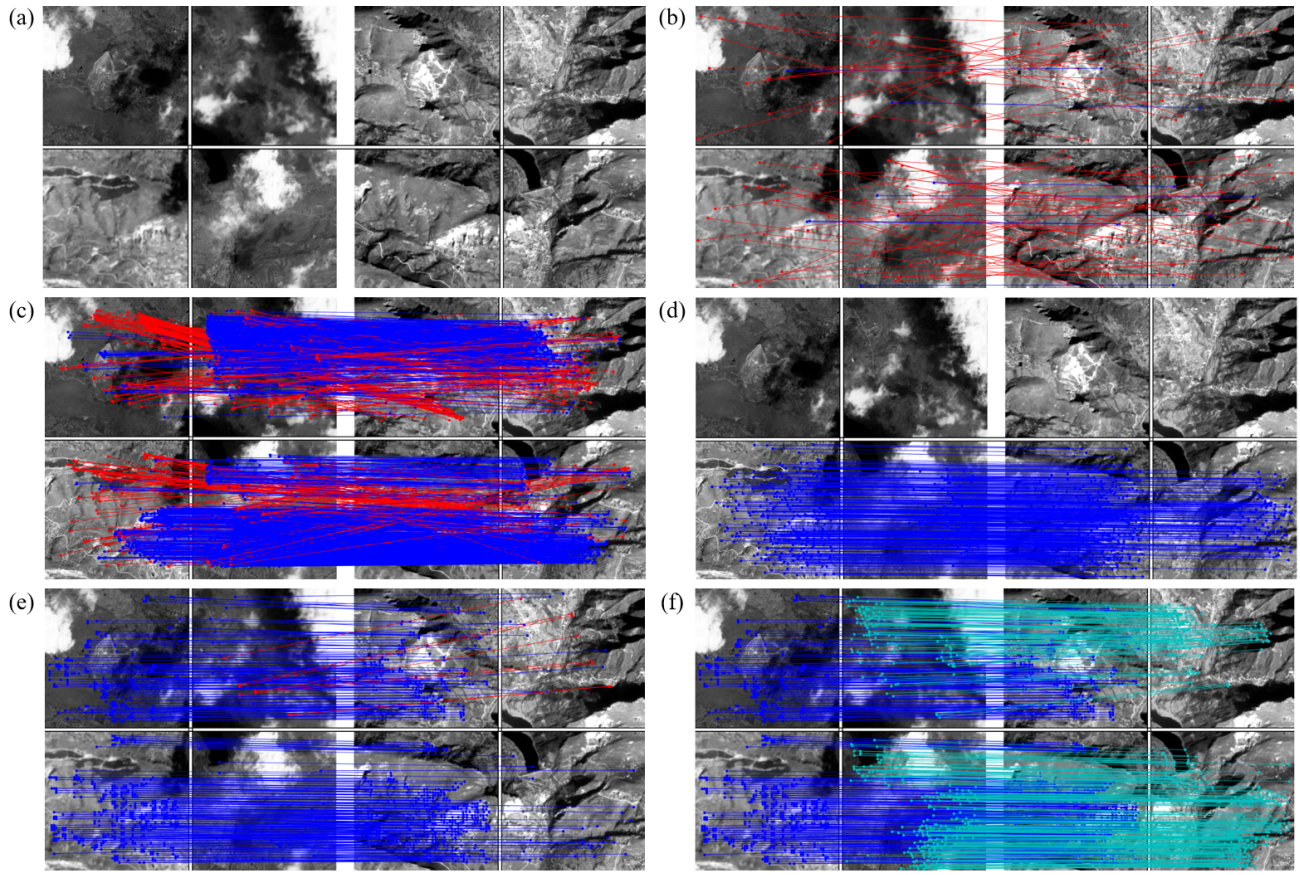

Fig. 14. Comparison of feature matching results in local regions. (a) Original image. (b) SIFT. (c) RIFT. (d) CoFSM. (e) SP + SG. (f) Proposed method.

TABLE V
COMPARISON OF BUNDLE ADJUSTMENT METHODS ON EARTH OBSERVATION DATA

Method	**GF-7 Residual (pixels)**			**ZY-3 Residual (pixels)**		
	x	y	xy	x	y	xy
Standard Bundle Adjustment	1.45	1.23	1.90	2.18	1.96	2.93
Weighted Bundle Adjustment	**1.27**	**0.98**	**1.60**	**2.11**	**1.91**	**2.85**

TABLE VI
COMPARISON OF BUNDLE ADJUSTMENT METHODS ON MARS OBSERVATION DATA

Method	**HiRISE Residual (pixels)**			**CTX Residual (pixels)**		
	x	y	xy	x	y	xy
Standard Bundle Adjustment	1.07	3.24	3.41	0.98	2.14	2.35
Weighted Bundle Adjustment	**1.02**	**1.19**	**1.57**	**0.93**	**2.02**	**2.22**

image registration, 3-D reconstruction, and change detection in remote sensing. Future research may extend this methodology to other types of satellite imagery or different remote sensing modalities and integrate it with downstream tasks such as dense mapping and semantic analysis to further enhance the capabilities of remote sensing image processing.

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